

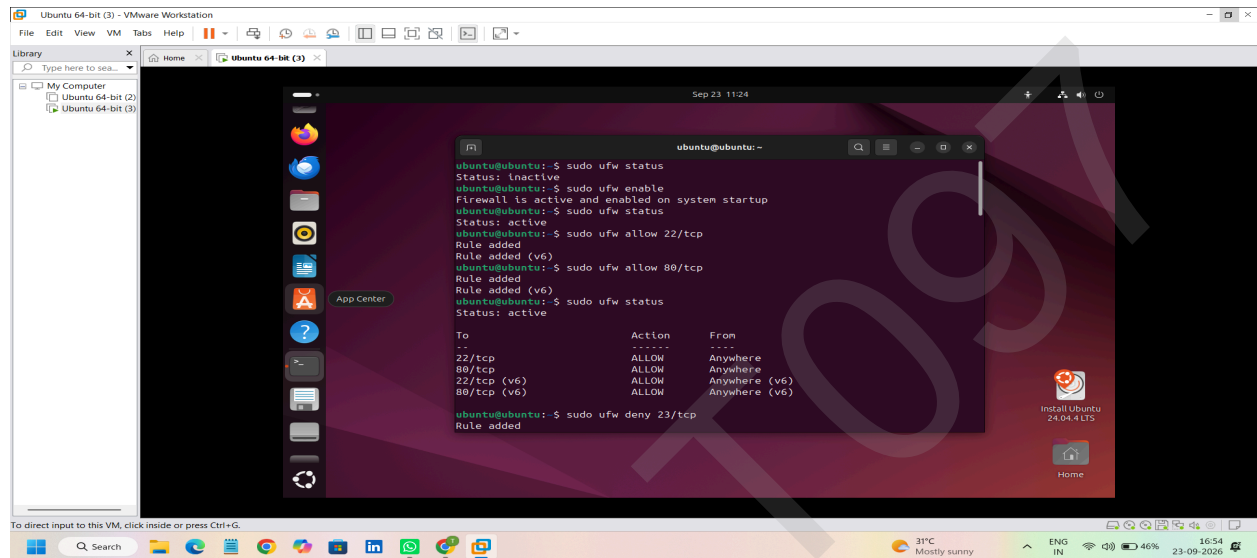
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Practical No 10

Aim: Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

Step 1: Open Ubuntu Terminal

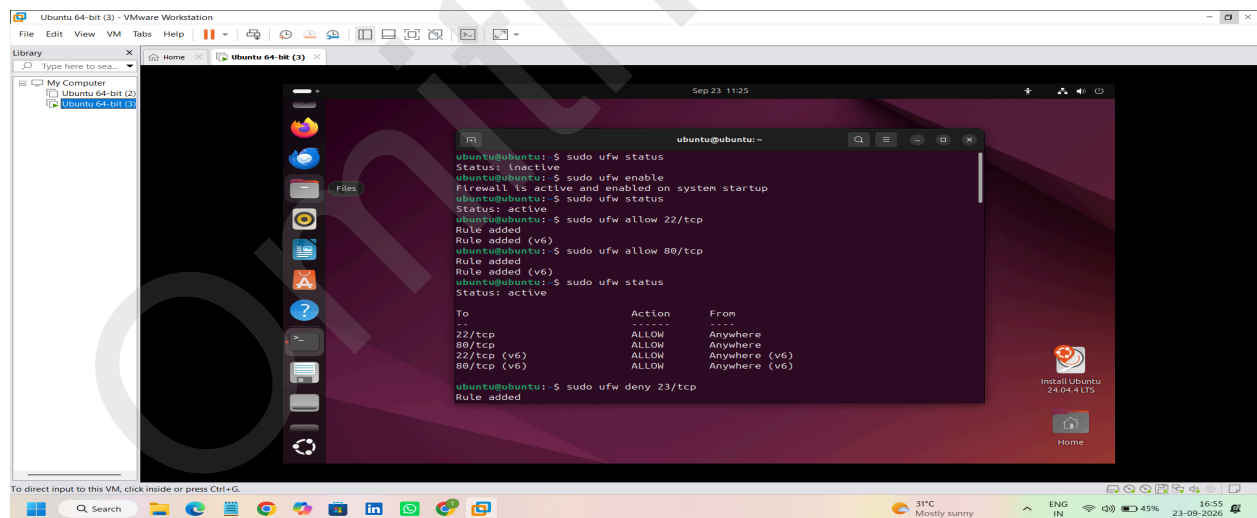


```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

Step 2: Enable the Firewall



```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

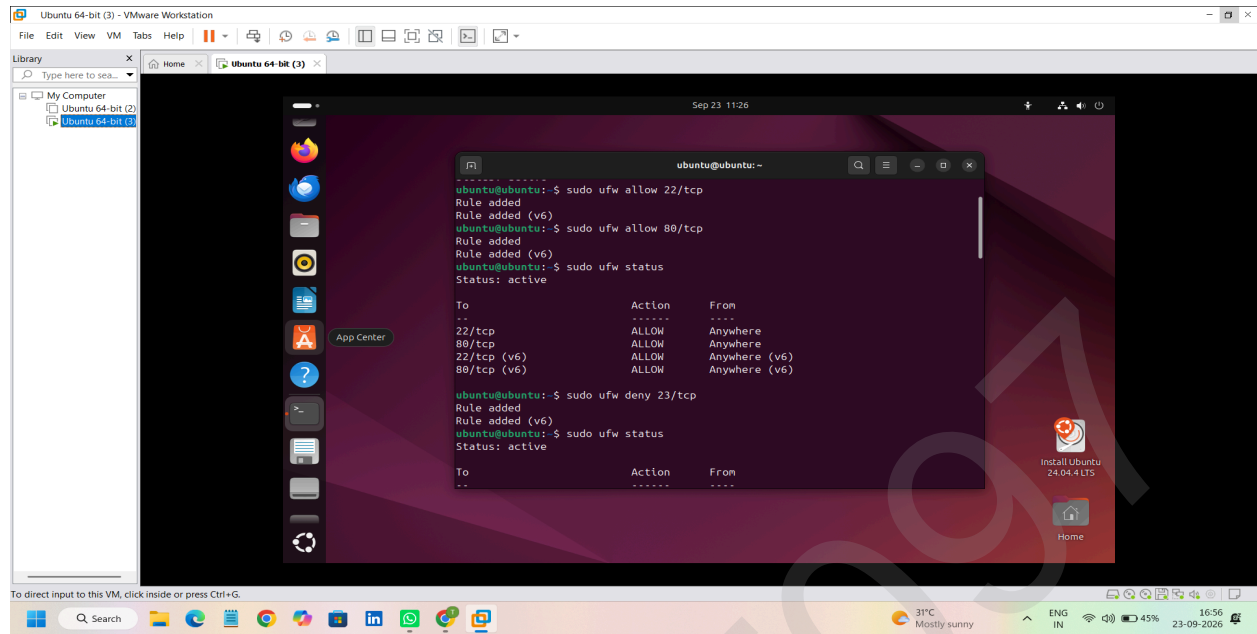
To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

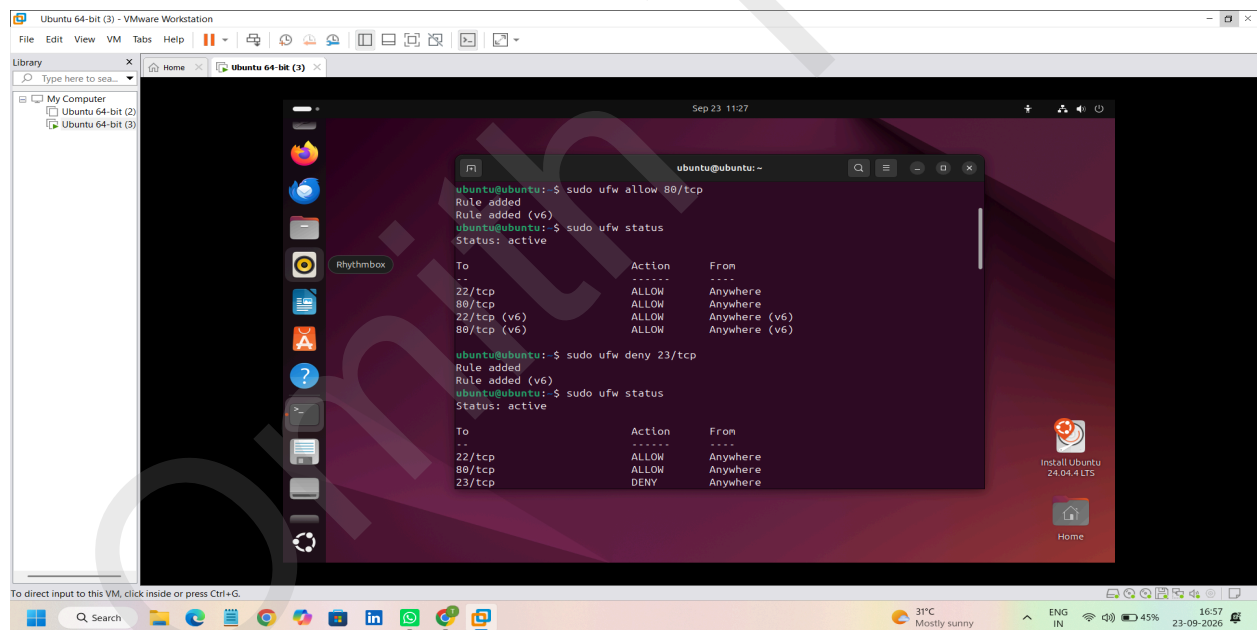
Step 3: Allow SSH Traffic

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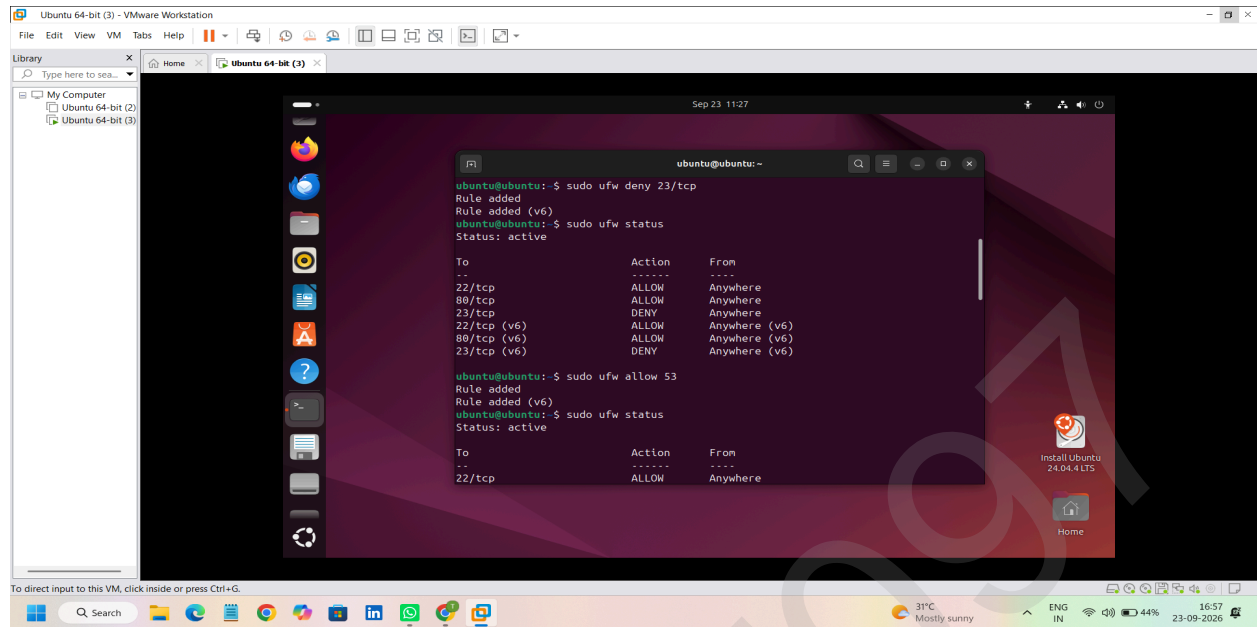
Step 4:- Step 4: Allow HTTP Traffic



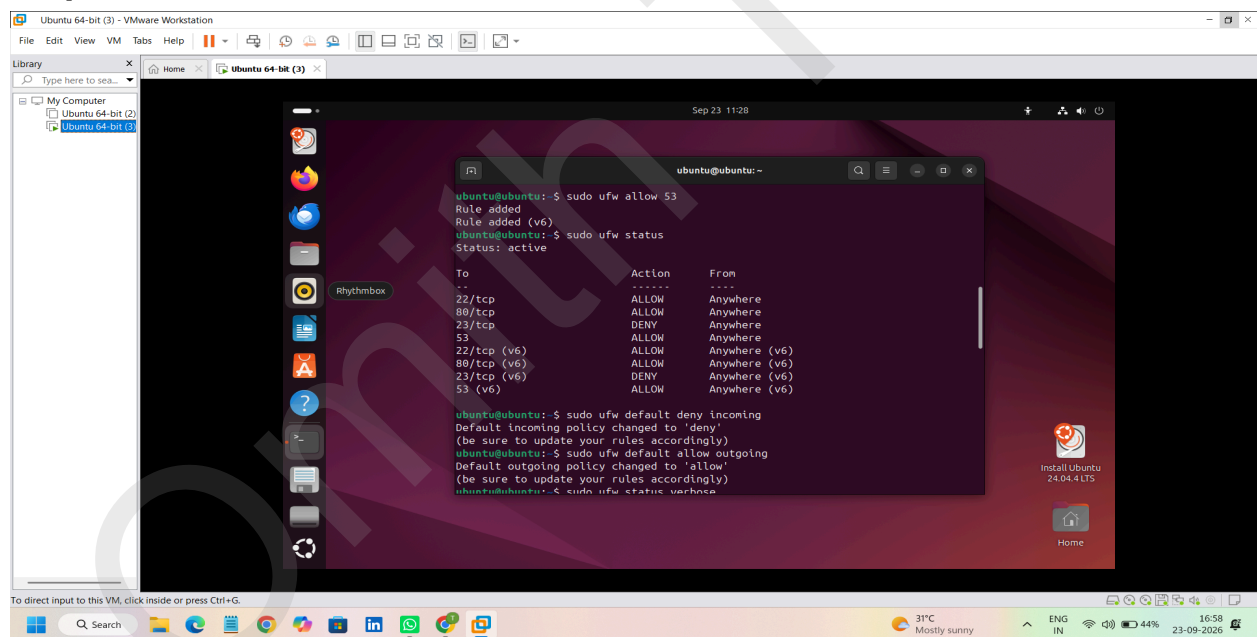
Step 5: Block a Port

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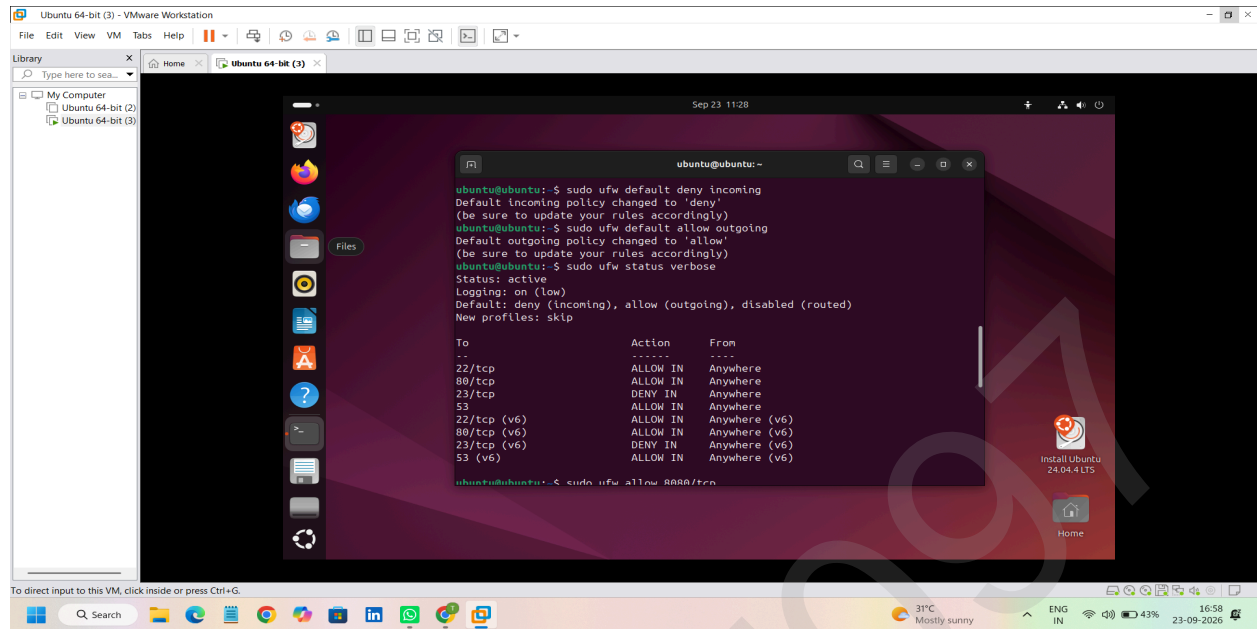
Step 6: Allow DNS Traffic



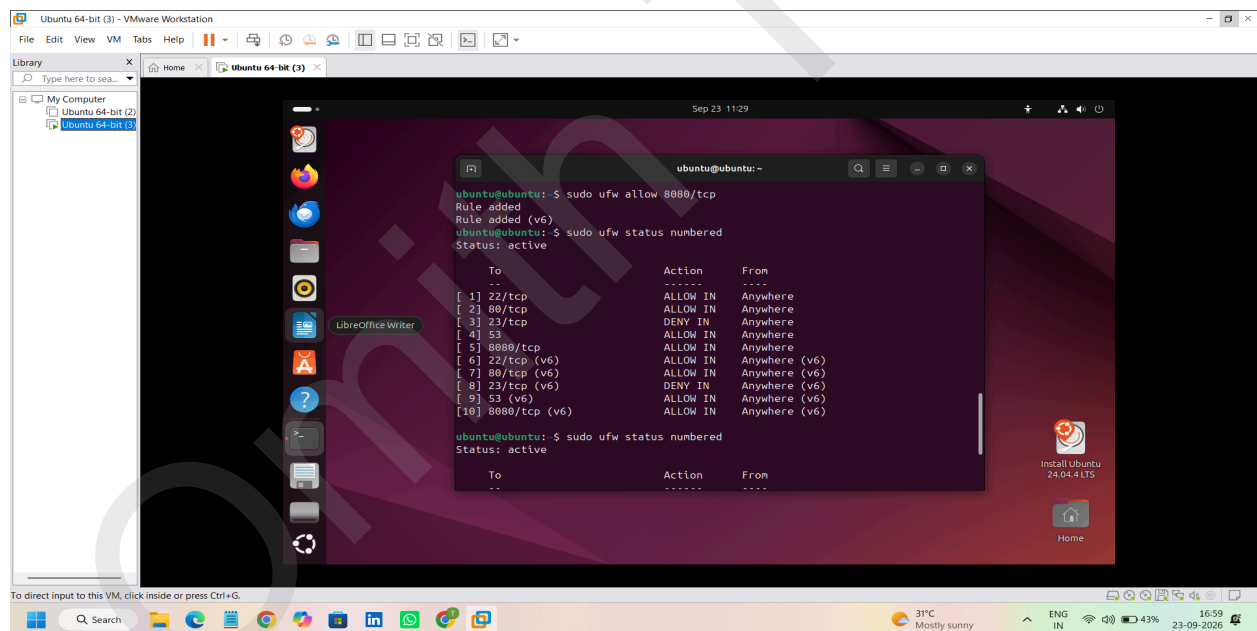
Step 7: Set Default Firewall Policies

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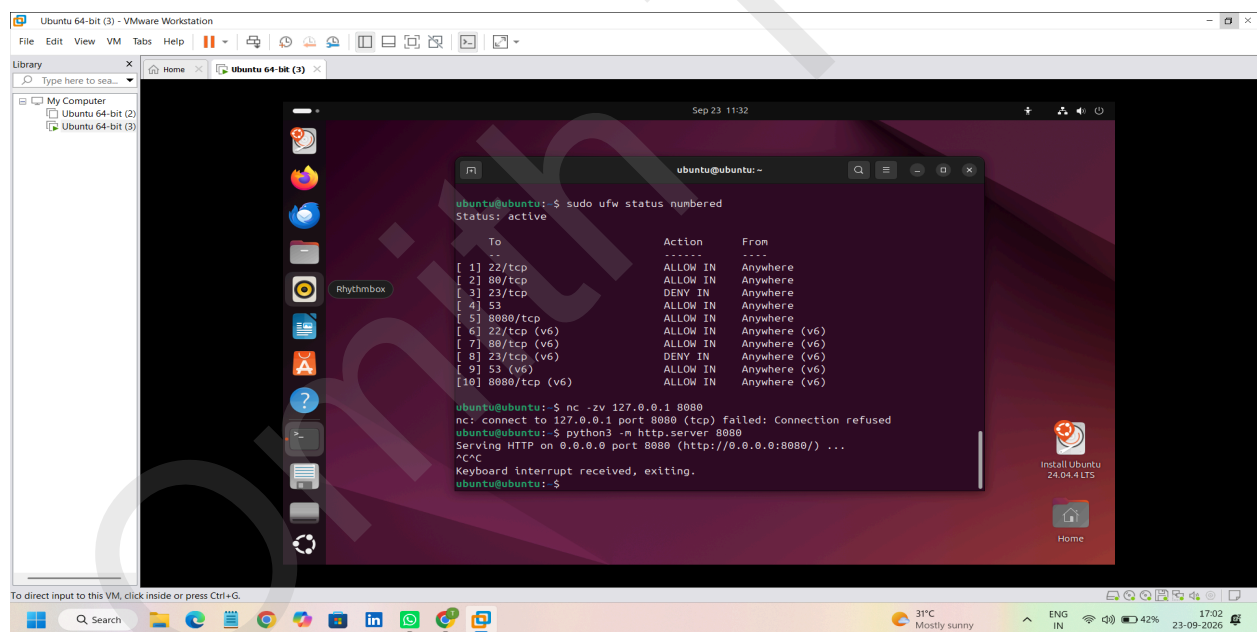
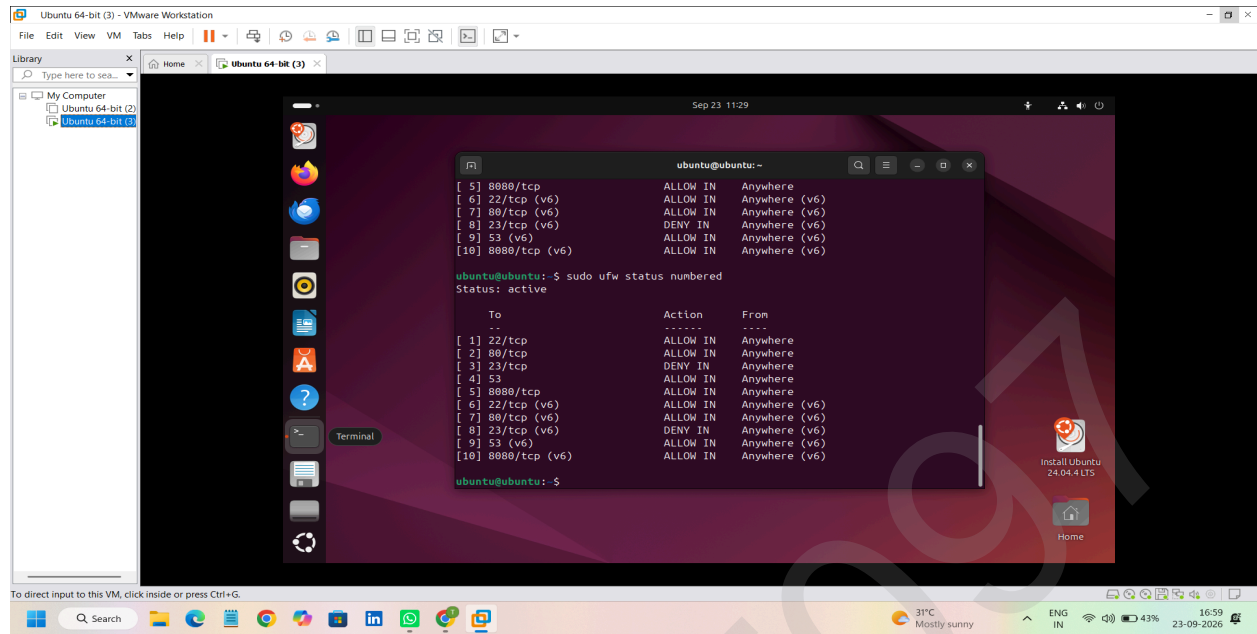
Step 8: Create a Rule for a Specific Port



Step 9: Test the Firewall

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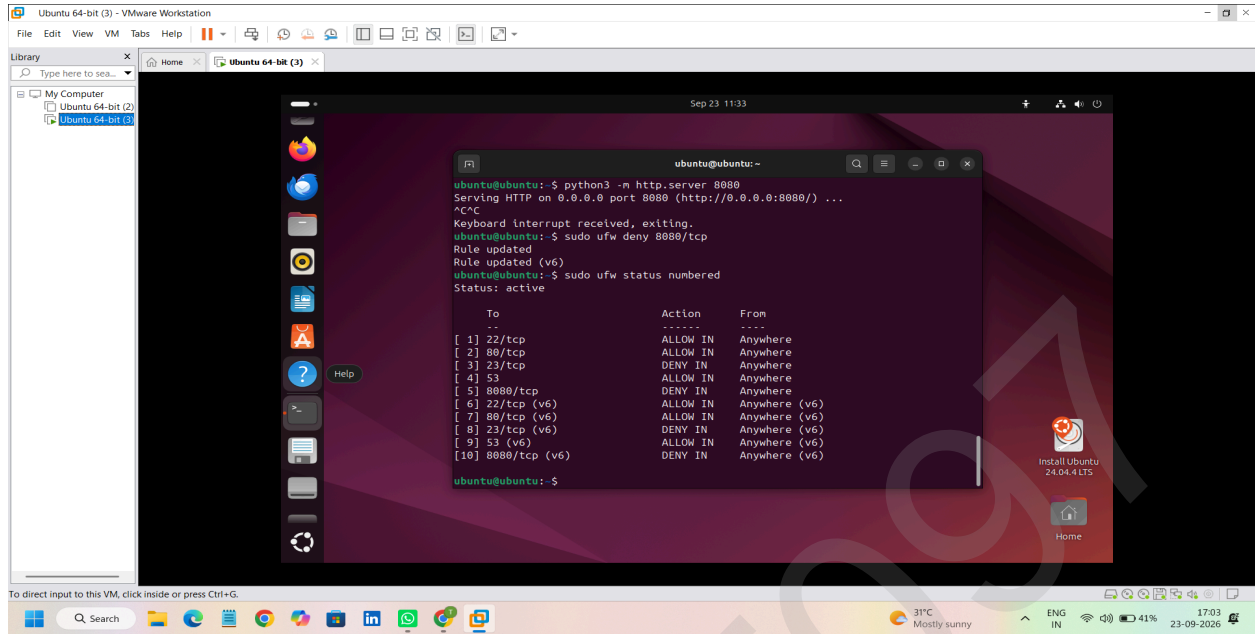
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Step 10: Test Blocking

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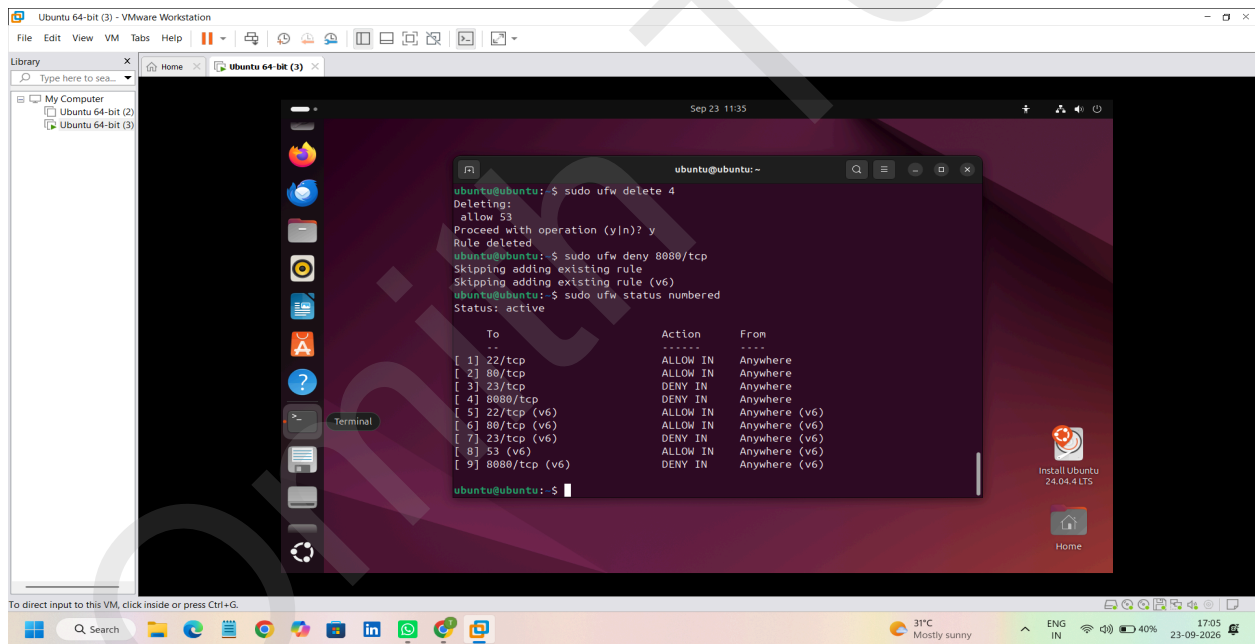
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```
ubuntu@ubuntu:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
^C
Keyboard Interrupt received, exiting.
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp
Rule updated
Rule updated (v6)
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To           Action      From
--           -
[ 1] 22/tcp   ALLOW IN    Anywhere
[ 2] 80/tcp   ALLOW IN    Anywhere
[ 3] 23/tcp   DENY IN     Anywhere
[ 4] 53        ALLOW IN    Anywhere
[ 5] 8080/tcp  DENY IN     Anywhere
[ 6] 22/tcp (v6) ALLOW IN    Anywhere (v6)
[ 7] 80/tcp (v6) ALLOW IN    Anywhere (v6)
[ 8] 23/tcp (v6) DENY IN     Anywhere (v6)
[ 9] 53 (v6)  ALLOW IN    Anywhere (v6)
[10] 8080/tcp (v6) DENY IN     Anywhere (v6)

ubuntu@ubuntu:~$
```



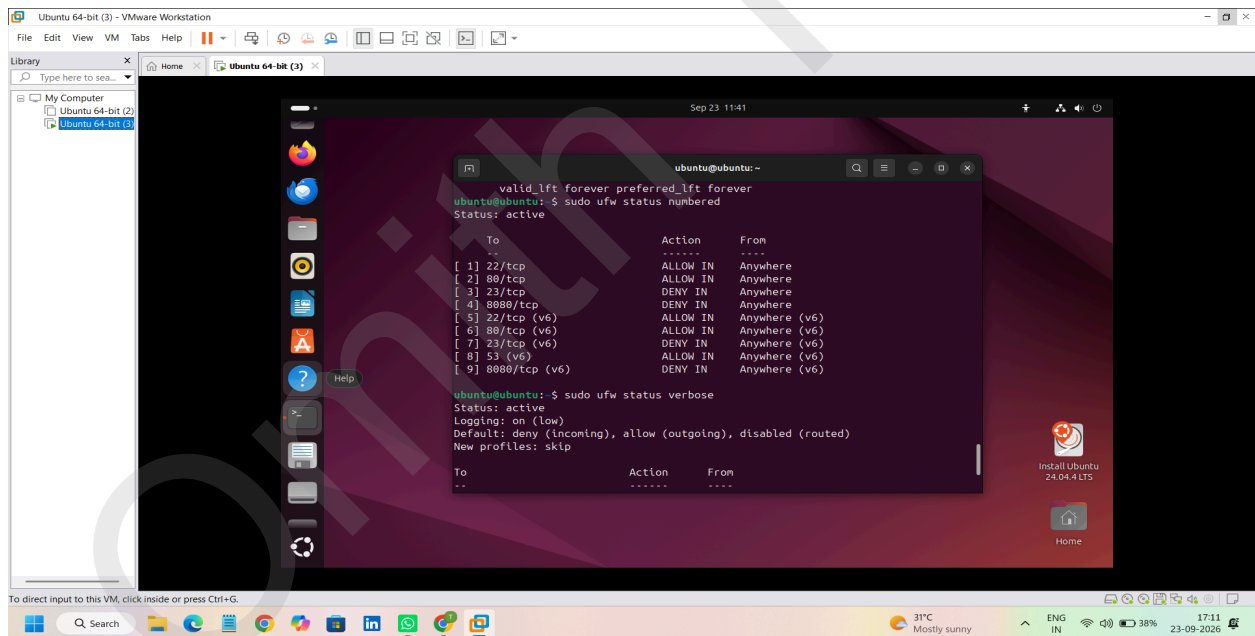
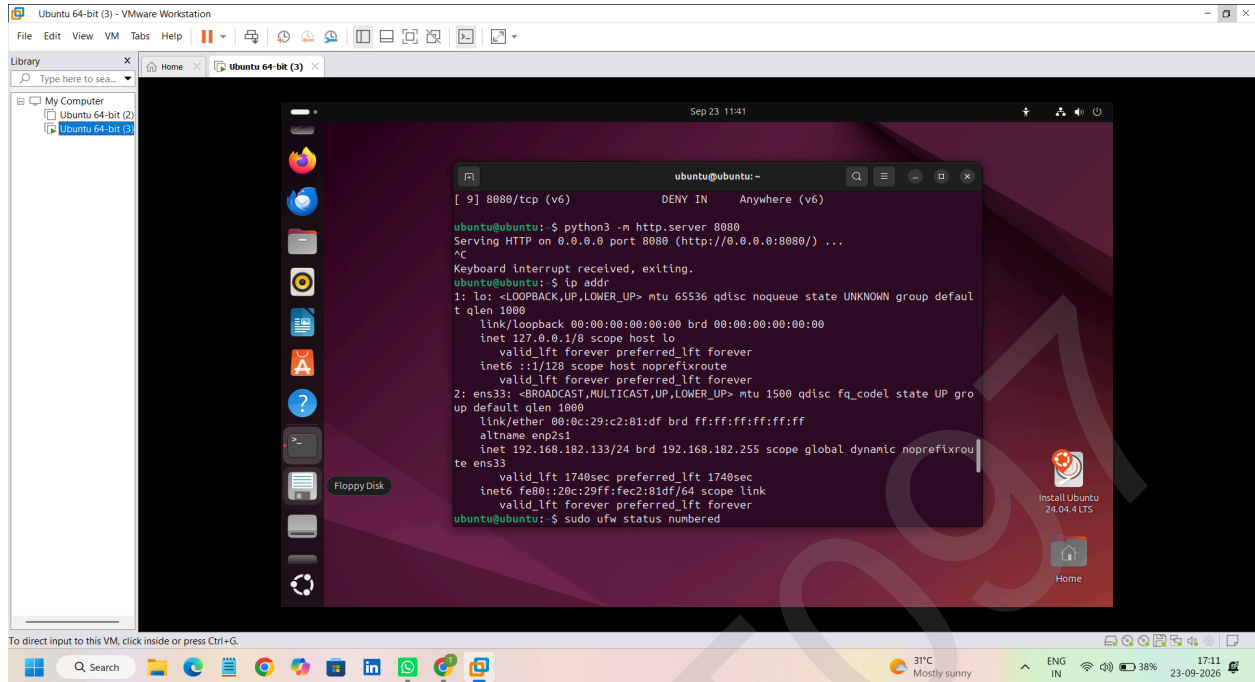
```
ubuntu@ubuntu:~$ sudo ufw delete 4
Deleting:
allow 53
Proceed with operation (y/n)? y
Rule deleted
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To           Action      From
--           -
[ 1] 22/tcp   ALLOW IN    Anywhere
[ 2] 80/tcp   ALLOW IN    Anywhere
[ 3] 23/tcp   DENY IN     Anywhere
[ 4] 8080/tcp  DENY IN     Anywhere
[ 5] 22/tcp (v6) ALLOW IN    Anywhere (v6)
[ 6] 80/tcp (v6) ALLOW IN    Anywhere (v6)
[ 7] 23/tcp (v6) DENY IN     Anywhere (v6)
[ 8] 53 (v6)  ALLOW IN    Anywhere (v6)
[ 9] 8080/tcp (v6) DENY IN     Anywhere (v6)

ubuntu@ubuntu:~$
```

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